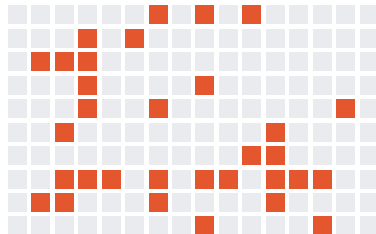


MASTER'S THESIS · BOCCONI UNIVERSITY

Imputation of Missing Data in Financial Time Series

From reconstruction error to downstream decisions



CANDIDATE

Michele Carlo Fanelli

SUPERVISOR

Francesco Corielli

ACADEMIC YEAR

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Financial data is never fully observed

Where data goes missing

- Order-book feeds drop quotes in bursts of cancellations
- Trading halts when prices move too violently
- OTC bond tenors go unquoted for days
- Macro data arrive at mixed frequencies, with lags

The usual fixes distort risk

- Carrying the last price forward: stale prices bias betas (Scholes and Williams, 1977)
- Dropping incomplete dates: unbiased only under MCAR, and much of the sample is lost
- Filling with a constant: variances and correlations are damped, even under MCAR

What is left open

- Deep imputers assume MAR, usually implicitly through their masked training objective
- Evaluation is almost always pointwise, RMSE or MAE on masked cells, which says little about the covariance structure

Aim of the thesis. Evaluate each imputation on its reconstruction accuracy and on the downstream model built on it, against the same model fitted on the complete panel.

Four research questions

1. Mechanism or amount?

What drives the damage, the share of missing data or the mechanism that produced it, at comparable shares?

2. Do the deep imputers earn their cost?

Recurrent and diffusion models against classical likelihood and low-rank estimators, with training and sampling time counted.

3. Is pointwise accuracy a good proxy?

Does a method that fills the holes accurately also preserve covariances, tails, marginals and temporal dependence?

4. Do the rankings survive downstream?

Is the ordering given by reconstruction metrics the ordering given by decisions taken on the imputed panel?

CONTRIBUTIONS

Controlled testbed

Fifty S&P 500 stocks, three mechanisms at five severities, seven imputers in one framework.

Evaluation suite

Past reconstruction error: covariance, Value-at-Risk, Wasserstein, volatility clustering and leverage.

Downstream evaluations

Tangency portfolio and factor structure, ranking imputers against each other and an oracle.

A controlled experiment on a complete equity panel

1 Complete panel

S&P 500 end-of-day prices, 1/1/2010 to 30/5/2026. Stocks with any missing price removed; 50 sampled across industries.

4125 × 50 panel



2 Inject missingness

Three mechanisms, two MCAR and one MNAR, five levels each, matched on the expected share of missing cells.

15 corrupted panels



3 Impute

Seven methods, from deletion to conditional diffusion, run on every corrupted panel.

105 runs (7 × 15)



4 Evaluate

Seven metrics against the true panel; two downstream models against an oracle fitted on it.

7 metrics + 2 models

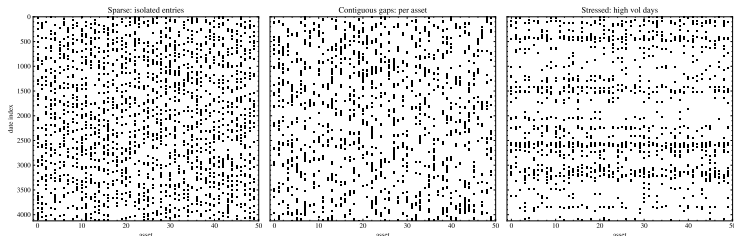
Returns, not prices

Prices are positive, level dependent and non-stationary, so the panel is in daily log returns. A deleted return is a stylised missing price: a carried-forward price is a zero return, but the later catch-up return is not modelled.

Offline reconstruction

Every method may use the full sample, including dates after the gap. The results describe reconstruction after the fact, not real-time filling.

Three mechanisms, matched on the share of missing cells



Sparse (MCAR)

Each cell is deleted independently with probability r .

Contiguous gaps (MCAR)

n seed cells, each extended forward into a run with mean length ℓ trading days.

Stressed (MNAR)

Dates at or above the p -quantile of the squared panel-average return lose each entry with probability h , other dates with r . The score is hidden from the imputer.

Expected share of missing cells		0.1%	1%	5%	10%	30%
Sparse	r	0.001	0.01	0.05	0.1	0.3
Gaps	(n, ℓ)	(20, 10)	(200, 10)	(1000, 10)	(1000, 20)	(2000, 30)
Stressed	(r, h, p)	(0, 0.1, 0.99)	(0.005, 0.5, 0.99)	(0.04, 0.2, 0.95)	(0.05, 0.5, 0.9)	(0.2, 1, 0.9)

Seven methods, from deletion to conditional diffusion

Baselines	Deletion	Drop every date with a missing value (complete cases)	Little and Rubin (2019)
	Zero	Fill with 0, as if the price were carried forward	
Classical	KNN	Inverse-distance average over the k nearest dates	Troyanskaya et al. (2001)
	Low-rank	Iterative truncated SVD: a q -factor reconstruction	Mazumder et al. (2010)
	EM	Gaussian maximum likelihood, conditional-mean fill	Dempster et al. (1977)
Deep learning	BRITS	Bidirectional RNN blending history and cross-section	Cao et al. (2018)
	CSDI	Conditional score-based diffusion; median of 32 draws	Tashiro et al. (2021)

$$\mathbb{E}[r_{t,m} \mid r_{t,o}] = \mu_m + \Sigma_{mo} \Sigma_{oo}^{-1} (r_{t,o} - \mu_o)$$

EM fill: the conditional mean given the assets observed on the same date, which minimises the expected squared error

Evaluating the panel, not only the filled cells

Seven metrics, four families

Pointwise	MAE, RMSE	Error on the imputed cells only
Marginal	VaRErr	5% Value-at-Risk, per asset
	WassErr	Wasserstein distance of the marginals
Second moment	CovErr	$\ \hat{\Sigma} - \Sigma\ _F / \ \Sigma\ _F$
Temporal	AbsACFErr	Volatility clustering, 20 lags
	LevErr	Leverage effect, $\text{corr}(r_t, r_{t+k})$

The first five are invariant to permuting the dates; the temporal ones are not.

Two downstream evaluations

Tangency portfolio

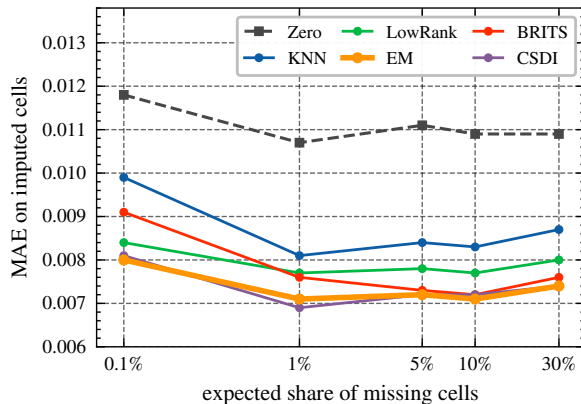
$\hat{w} = \frac{\hat{\Sigma}^{-1} \hat{\mu}}{\mathbf{1}^\top \hat{\Sigma}^{-1} \hat{\mu}}$ fitted on each panel, evaluated on the true returns, compared with the oracle fitted on the complete panel (in sample).

Factor structure

Variance shares $\hat{\lambda}_i / \sum_j \hat{\lambda}_j$ and subspace distance

$D_k = \frac{1}{\sqrt{k}} \left(\sum_{i=1}^k \sin^2 \theta_i \right)^{1/2} \in [0, 1]$ between the true and estimated leading eigenspaces.

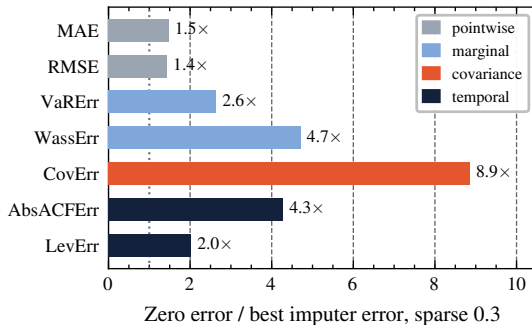
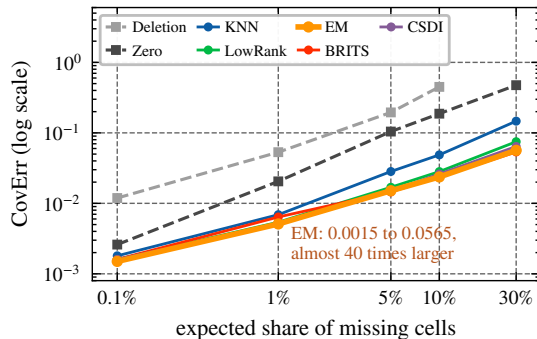
Pointwise errors hit a floor at once, and EM sits on it



Sparse pattern, MAE on the imputed cells (Table 3.1).

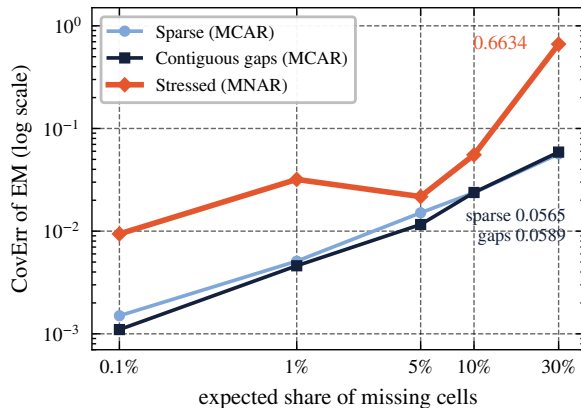
- Daily returns are close to serially uncorrelated: what can be known about a missing entry is carried by the other assets on the same date.
- EM's conditional mean extracts exactly that. Its MAE stays between **0.0071** and **0.0080** across all sparse levels.
- The five conditional methods are separated in the third decimal; the extra capacity of BRITS and CSDI has little to act on.
- The ranking is stable across patterns: EM is rarely beaten, and never by a wide margin.

Accurate cells do not make an accurate panel



- MAE averages over the filled cells, CovErr accumulates them over the panel: EM's CovErr grows almost 40-fold while its MAE stays flat.
- At sparse 0.3, Zero's MAE is **1.5 times** the best method's and its CovErr **8.9 times** (ratios of Table 3.1 values).
- The best fill cell by cell is a conditional mean, less variable than the return it replaces: Low-rank loses to EM on MAE but beats it on CovErr at two stressed levels.

The mechanism matters more than the amount

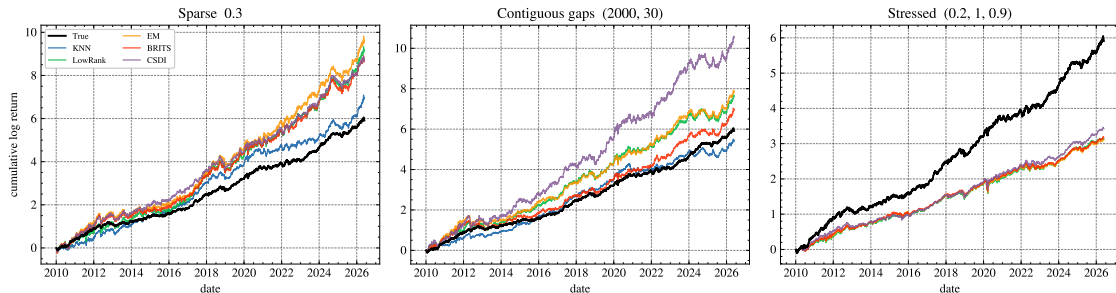


EM, covariance error by pattern at matched shares (Tables 3.1 to 3.3).

- The two MCAR patterns are nearly interchangeable: with 49 other assets observed on each date, breaking the time axis costs a cross-sectional imputer little.
- Stressed deletions are aligned by date. They remove the conditioning set, and what they remove are the largest returns.
- BRITS falls behind where deletions are most extreme: MAE **0.0301** against EM's 0.0133 at (0, 0.1, 0.99).
- At (0.2, 1, 0.9) the ranking compresses: CovErr between 0.648 and 0.663 for every surviving model.

EM CovErr	Sparse	Gaps	Stressed
about 10% missing	0.0237	0.0238	0.0554
about 30% missing	0.0565	0.0589	0.6634

Downstream I: the portfolio amplifies the distortion



Oracle

complete panel, annualised mean 0.361, variance 0.049
Sharpe **1.636**

Heavy MCAR

Variance rises from 0.049 to as much as 0.29 and L^1 weight distances reach 14: the inverted $\hat{\Sigma}$ over-leverages.

Heavy MNAR

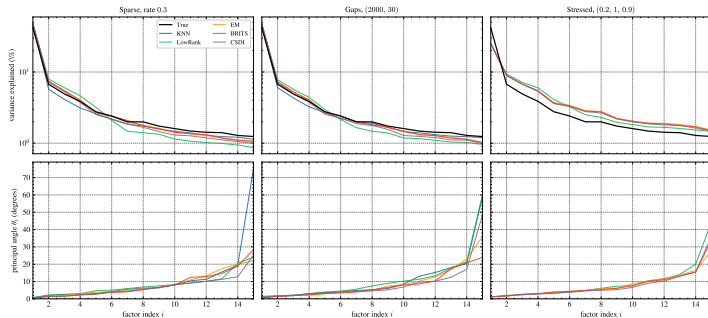
The imputers never see the volatility; their portfolios end at little more than half of the oracle's cumulative return.

Who fails

EM and Low-rank are never the worst. BRITS: Sharpe 1.30 against EM's 1.51 at (0.05, 0.5, 0.9).

In-sample diagnostic of reconstruction fidelity, not investment performance. Deletion: Sharpe near zero at sparse 0.1; its covariance is singular at the heavy levels.

Downstream II: a robust market factor, a fragile subspace



Top: variance share of the first 15 components (log scale). Bottom: ordered principal angles between the 15-factor subspaces, not per-factor errors. Heaviest level of each pattern.

- Market direction: $D_1 < 0.026$ at sparse 0.3, under 1.5 degrees
- Five-factor subspace: D_5 from 0.14 to 0.39, six to eighteen times larger
- Conditional fills inflate the first factor: **41.74%** to between 45.6% and 48.5% at sparse 0.3
- Low-rank has the largest D_5 at 10 of 15 levels, although it imposes a factor structure
- At (0.2, 1, 0.9) the first factor falls to about 25% for all four methods that can be evaluated

$\hat{\Sigma}^{-1} = \sum_i \hat{\lambda}_i^{-1} \hat{v}_i \hat{v}_i^\top$: the portfolio loads most on the trailing directions, the ones the imputations recover worst. Its error maximisation is the factor distortion seen through an inverse.

The deep models did not earn their cost

1 s

EM

about one second to fill
a 4125×50 panel

1 h

BRITS

just under an hour of
training per panel
(10^4 epochs)

4 h

CSDI

just under four hours of
training per panel, then
about 24 minutes of
sampling

70 h

All neural training

roughly seventy hours
for the 15 levels, four
fifths of it on CSDI

CSDI matches EM's accuracy at about **four orders of magnitude** more computation, and BRITS falls behind at two stressed levels.

Single Apple Mac mini (M4, 16 GB), CPU-only PyTorch. Small networks: about 1.2×10^5 parameters for BRITS and 4.7×10^4 for CSDI. Times are approximate, as reported in Section 3.3 of the thesis.

Answers to the four questions

1 Mechanism or amount?

The mechanism, in this experiment. Under MCAR the methods hold with 30% missing; under stressed MNAR they break at far smaller shares, and at the heaviest level no method separates.

2 Do the deep imputers earn their cost?

Not here. EM is rarely beaten; CSDI matches it at four orders of magnitude more cost; BRITS falls behind at two stressed levels.

3 Is pointwise accuracy a good proxy?

Poor. Zero's MAE is 1.5 times the best and its CovErr about 9 times. Conditional fills inflate the first factor, from 41.7% to as much as 48.5%, which no pointwise metric registers.

4 Do the rankings survive downstream?

Largely, and the exceptions are informative. Low-rank: accurate cells, worst five-factor subspace at 10 of 15 levels. Deletion: inefficient in the tables, unstable portfolios downstream.

Score an imputation against the use it is put to.

Limitations and future work

Limitations

- One panel: fifty liquid large caps, daily, one period, one run per configuration
- Injected missingness; the stressed pattern is a stylised MNAR that mixes the selection rule with date-aligned deletion
- Gaussian EM without the Student- t refinement; both downstream models read $\hat{\Sigma}$; the portfolio is in sample
- Offline setting: BRITS and CSDI use observations after the gap

Future work

- Causal imputation with only past data, to measure the cost of filling in real time
- Downstream tests beyond the second moment: VaR backtests, hedging, trading rules
- Multiple imputation: let the decision see the spread of the CSDI draws
- Larger cross sections, other frequencies and asset classes, real gaps

Thank you

Backup: evaluation scores under sparse missingness (Table 3.1; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
0.001	Deletion	–	–	0.0119	0.0001	0.0001	0.0101	0.0114
	Zero	0.0118	0.0194	0.0026	0.0000	0.0000	0.0009	0.0008
	KNN	0.0099	0.0168	0.0018	0.0000	0.0000	0.0006	0.0005
	LowRank	0.0084	0.0152	0.0016	0.0000	0.0000	0.0005	0.0004
	EM	0.0080	0.0147	0.0015	0.0000	0.0000	0.0005	0.0004
	BRITS	0.0091	0.0158	0.0016	0.0000	0.0000	0.0005	0.0004
	CSDI	0.0081	0.0151	0.0016	0.0000	0.0000	0.0005	0.0004
0.01	Deletion	–	–	0.0532	0.0005	0.0004	0.0247	0.0334
	Zero	0.0107	0.0161	0.0205	0.0001	0.0001	0.0035	0.0027
	KNN	0.0081	0.0123	0.0069	0.0001	0.0000	0.0019	0.0017
	LowRank	0.0077	0.0117	0.0054	0.0000	0.0000	0.0016	0.0015
	EM	0.0071	0.0109	0.0051	0.0000	0.0000	0.0015	0.0014
	BRITS	0.0076	0.0116	0.0064	0.0001	0.0000	0.0017	0.0016
	CSDI	0.0069	0.0107	0.0050	0.0000	0.0000	0.0015	0.0014
0.05	Deletion	–	–	0.1951	0.0019	0.0012	0.1334	0.0515
	Zero	0.0111	0.0167	0.1044	0.0005	0.0006	0.0132	0.0073
	KNN	0.0084	0.0127	0.0283	0.0003	0.0002	0.0047	0.0045
	LowRank	0.0078	0.0118	0.0169	0.0002	0.0001	0.0045	0.0041
	EM	0.0072	0.0113	0.0151	0.0002	0.0002	0.0044	0.0039
	BRITS	0.0073	0.0113	0.0145	0.0002	0.0002	0.0043	0.0040
	CSDI	0.0072	0.0112	0.0149	0.0002	0.0002	0.0051	0.0041
0.1	Deletion	–	–	0.4492	0.0091	0.0051	–	–
	Zero	0.0109	0.0163	0.1872	0.0011	0.0011	0.0276	0.0111
	KNN	0.0083	0.0125	0.0487	0.0007	0.0004	0.0090	0.0067
	LowRank	0.0077	0.0116	0.0282	0.0004	0.0003	0.0072	0.0058
	EM	0.0071	0.0110	0.0237	0.0005	0.0003	0.0068	0.0056
	BRITS	0.0072	0.0111	0.0238	0.0005	0.0003	0.0068	0.0057
	CSDI	0.0072	0.0111	0.0261	0.0004	0.0003	0.0077	0.0057
0.3	Deletion	–	–	–	–	–	–	–
	Zero	0.0109	0.0161	0.4749	0.0037	0.0033	0.0638	0.0201
	KNN	0.0087	0.0129	0.1467	0.0024	0.0014	0.0173	0.0118
	LowRank	0.0080	0.0119	0.0749	0.0014	0.0007	0.0170	0.0105
	EM	0.0074	0.0112	0.0565	0.0017	0.0009	0.0150	0.0099
	BRITS	0.0076	0.0114	0.0536	0.0015	0.0008	0.0149	0.0101
	CSDI	0.0074	0.0113	0.0641	0.0014	0.0008	0.0203	0.0112

Backup: evaluation scores under contiguous-gap missingness (Table 3.2; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(20, 10)	Deletion	—	—	0.0103	0.0001	0.0001	0.0033	0.0033
	Zero	0.0121	0.0184	0.0027	0.0000	0.0000	0.0003	0.0002
	KNN	0.0096	0.0144	0.0016	0.0000	0.0000	0.0002	0.0001
	LowRank	0.0087	0.0123	0.0010	0.0000	0.0000	0.0002	0.0001
	EM	0.0079	0.0117	0.0011	0.0000	0.0000	0.0001	0.0001
	BRITS	0.0078	0.0112	0.0010	0.0000	0.0000	0.0001	0.0001
	CSDI	0.0079	0.0115	0.0009	0.0000	0.0000	0.0001	0.0001
(200, 10)	Deletion	—	—	0.1492	0.0010	0.0004	0.0243	0.0158
	Zero	0.0107	0.0156	0.0222	0.0001	0.0001	0.0037	0.0015
	KNN	0.0082	0.0121	0.0065	0.0001	0.0000	0.0016	0.0012
	LowRank	0.0078	0.0118	0.0061	0.0001	0.0000	0.0015	0.0012
	EM	0.0072	0.0109	0.0046	0.0001	0.0000	0.0014	0.0011
	BRITS	0.0073	0.0111	0.0049	0.0001	0.0000	0.0015	0.0011
	CSDI	0.0072	0.0109	0.0049	0.0001	0.0000	0.0015	0.0011
(1000, 10)	Deletion	—	—	0.1825	0.0017	0.0010	0.1135	0.0493
	Zero	0.0104	0.0152	0.0785	0.0004	0.0005	0.0155	0.0048
	KNN	0.0080	0.0118	0.0224	0.0003	0.0002	0.0059	0.0032
	LowRank	0.0074	0.0108	0.0137	0.0002	0.0001	0.0048	0.0029
	EM	0.0069	0.0103	0.0116	0.0002	0.0001	0.0049	0.0028
	BRITS	0.0070	0.0104	0.0116	0.0002	0.0001	0.0049	0.0028
	CSDI	0.0069	0.0103	0.0114	0.0002	0.0001	0.0051	0.0027
(1000, 20)	Deletion	—	—	0.8524	0.0087	0.0042	0.1872	0.1681
	Zero	0.0110	0.0163	0.1716	0.0010	0.0010	0.0354	0.0068
	KNN	0.0085	0.0127	0.0487	0.0007	0.0004	0.0117	0.0049
	LowRank	0.0078	0.0119	0.0310	0.0004	0.0003	0.0088	0.0046
	EM	0.0073	0.0114	0.0238	0.0005	0.0003	0.0083	0.0044
	BRITS	0.0074	0.0114	0.0241	0.0005	0.0003	0.0082	0.0045
	CSDI	0.0073	0.0115	0.0328	0.0004	0.0003	0.0095	0.0044
(2000, 30)	Deletion	—	—	1.4799	0.0104	0.0100	—	—
	Zero	0.0108	0.0161	0.4324	0.0028	0.0027	0.0908	0.0143
	KNN	0.0085	0.0128	0.1378	0.0018	0.0011	0.0277	0.0090
	LowRank	0.0080	0.0120	0.0820	0.0010	0.0006	0.0218	0.0083
	EM	0.0073	0.0113	0.0589	0.0012	0.0007	0.0212	0.0079
	BRITS	0.0074	0.0114	0.0555	0.0011	0.0006	0.0187	0.0082
	CSDI	0.0074	0.0115	0.0896	0.0010	0.0007	0.0265	0.0091

Backup: evaluation scores under stressed missingness (Table 3.3; best value per level in bold)

Level	Model	MAE	RMSE	CovErr	VaRErr	WassErr	AbsACFErr	LevErr
(0, 0.1, 0.99)	Deletion	–	–	0.2813	0.0009	0.0004	0.0582	0.0200
	Zero	0.0552	0.0640	0.0676	0.0001	0.0001	0.0115	0.0070
	KNN	0.0202	0.0285	0.0229	0.0000	0.0000	0.0034	0.0029
	LowRank	0.0158	0.0220	0.0116	0.0000	0.0000	0.0025	0.0020
	EM	0.0133	0.0187	0.0094	0.0000	0.0000	0.0021	0.0017
	BRITS	0.0301	0.0385	0.0253	0.0000	0.0000	0.0046	0.0036
	CSDI	0.0163	0.0226	0.0175	0.0000	0.0000	0.0027	0.0022
(0.005, 0.5, 0.99)	Deletion	–	–	0.2960	0.0010	0.0005	0.0687	0.0268
	Zero	0.0321	0.0463	0.2238	0.0005	0.0003	0.0483	0.0172
	KNN	0.0161	0.0261	0.0991	0.0001	0.0001	0.0168	0.0095
	LowRank	0.0129	0.0200	0.0279	0.0000	0.0001	0.0078	0.0067
	EM	0.0118	0.0189	0.0319	0.0000	0.0001	0.0077	0.0065
	BRITS	0.0122	0.0197	0.0401	0.0000	0.0001	0.0082	0.0069
	CSDI	0.0136	0.0233	0.0751	0.0000	0.0001	0.0132	0.0087
(0.04, 0.2, 0.95)	Deletion	–	–	0.5325	0.0043	0.0018	0.1079	0.0491
	Zero	0.0143	0.0219	0.2254	0.0010	0.0007	0.0375	0.0143
	KNN	0.0092	0.0141	0.0562	0.0003	0.0002	0.0092	0.0067
	LowRank	0.0083	0.0125	0.0253	0.0002	0.0001	0.0068	0.0052
	EM	0.0077	0.0117	0.0217	0.0002	0.0001	0.0062	0.0048
	BRITS	0.0078	0.0118	0.0231	0.0002	0.0001	0.0064	0.0049
	CSDI	0.0078	0.0122	0.0374	0.0002	0.0001	0.0078	0.0061
(0.05, 0.5, 0.9)	Deletion	–	–	0.6318	0.0056	0.0023	0.1230	0.0630
	Zero	0.0174	0.0249	0.5125	0.0030	0.0017	0.0843	0.0244
	KNN	0.0103	0.0161	0.1602	0.0010	0.0004	0.0258	0.0124
	LowRank	0.0091	0.0139	0.0483	0.0004	0.0002	0.0110	0.0091
	EM	0.0084	0.0130	0.0554	0.0008	0.0003	0.0107	0.0087
	BRITS	0.0106	0.0162	0.2083	0.0015	0.0006	0.0256	0.0126
	CSDI	0.0086	0.0137	0.0542	0.0008	0.0003	0.0165	0.0109
(0.2, 1, 0.9)	Deletion	–	–	–	–	–	–	–
	Zero	0.0149	0.0219	0.7677	0.0070	0.0042	0.1201	0.0333
	KNN	–	–	–	–	–	–	–
	LowRank	0.0137	0.0212	0.6525	0.0062	0.0029	0.1111	0.0316
	EM	0.0134	0.0210	0.6634	0.0065	0.0031	0.1112	0.0315
	BRITS	0.0136	0.0214	0.6480	0.0063	0.0025	0.1069	0.0299
	CSDI	0.0134	0.0210	0.6559	0.0064	0.0028	0.1065	0.0298

Backup: optimal portfolio at every level (Tables 3.4 to 3.6; best Sharpe in bold)

Expected share	0.1%	1%	Sparse			0.1%	Contiguous gaps				0.1%	1%	Stressed		
			5%	10%	30%		1%	5%	10%	30%			5%	10%	30%
Sharpe ratio (true 1.636)															
KNN	1.635	1.623	1.569	1.490	1.279	1.635	1.627	1.575	1.500	1.186	1.619	1.565	1.542	1.418	–
LowRank	1.635	1.624	1.565	1.529	1.253	1.635	1.627	1.593	1.560	1.225	1.629	1.606	1.586	1.494	1.195
EM	1.634	1.625	1.567	1.531	1.259	1.635	1.628	1.596	1.549	1.250	1.631	1.612	1.587	1.506	1.207
BRITS	1.634	1.625	1.572	1.542	1.226	1.635	1.629	1.588	1.555	1.276	1.620	1.602	1.593	1.304	1.212
CSDI	1.635	1.626	1.580	1.527	1.279	1.635	1.628	1.589	1.542	1.186	1.630	1.566	1.580	1.508	1.281
L^1 weight distance to the oracle															
KNN	0.30	1.11	2.92	4.07	8.43	0.16	0.87	2.24	4.31	6.52	1.13	2.50	3.16	4.64	–
LowRank	0.29	1.08	3.01	3.33	12.21	0.11	0.84	1.91	3.14	9.14	0.76	1.58	2.09	3.50	5.30
EM	0.30	1.03	3.06	3.40	13.76	0.10	0.84	1.77	3.40	10.17	0.60	1.44	2.00	3.45	5.22
BRITS	0.33	0.94	2.81	3.10	12.61	0.10	0.73	1.99	3.20	7.50	1.20	1.78	1.91	4.49	5.43
CSDI	0.30	1.02	2.89	3.03	11.33	0.11	0.79	1.97	3.52	14.48	0.66	2.91	2.62	3.50	5.22
Annualised mean (true 0.361)															
KNN	0.362	0.358	0.373	0.351	0.422	0.363	0.363	0.372	0.380	0.329	0.347	0.328	0.359	0.346	–
LowRank	0.362	0.360	0.391	0.375	0.556	0.362	0.365	0.369	0.386	0.465	0.356	0.353	0.365	0.359	0.191
EM	0.362	0.361	0.391	0.377	0.581	0.362	0.363	0.371	0.379	0.478	0.356	0.353	0.370	0.359	0.189
BRITS	0.362	0.359	0.384	0.371	0.528	0.362	0.362	0.369	0.358	0.423	0.345	0.349	0.369	0.243	0.193
CSDI	0.362	0.361	0.396	0.373	0.531	0.362	0.364	0.374	0.381	0.643	0.360	0.375	0.392	0.354	0.209
Annualised variance (true 0.0488)															
KNN	0.0491	0.0486	0.0564	0.0554	0.1086	0.0492	0.0497	0.0557	0.0644	0.0767	0.0459	0.0439	0.0543	0.0597	–
LowRank	0.0490	0.0492	0.0625	0.0603	0.1970	0.0491	0.0503	0.0537	0.0612	0.1439	0.0477	0.0483	0.0529	0.0576	0.0255
EM	0.0491	0.0493	0.0623	0.0607	0.2130	0.0491	0.0499	0.0539	0.0598	0.1460	0.0478	0.0480	0.0545	0.0568	0.0245
BRITS	0.0492	0.0487	0.0597	0.0577	0.1852	0.0490	0.0494	0.0540	0.0529	0.1098	0.0452	0.0476	0.0536	0.0347	0.0255
CSDI	0.0489	0.0492	0.0627	0.0597	0.1723	0.0489	0.0499	0.0555	0.0611	0.2937	0.0487	0.0572	0.0615	0.0551	0.0266

Backup: factor structure at every level (Tables 3.7 to 3.9; best D_5 in bold)

Expected share	0.1%	1%	Sparse			0.1%	Contiguous gaps				0.1%	1%	Stressed		
			5%	10%	30%		1%	5%	10%	30%			5%	10%	30%
Variance share of the first component (%) (true 41.74)															
KNN	41.76	41.87	42.28	43.00	45.62	41.75	41.90	42.29	42.93	44.63	41.31	39.81	41.68	40.27	–
LowRank	41.77	41.91	42.64	43.60	47.56	41.75	41.93	42.42	43.63	46.84	41.75	41.92	42.73	43.50	25.12
EM	41.77	41.91	42.64	43.59	47.88	41.75	41.92	42.42	43.52	46.95	41.73	41.80	42.66	43.29	24.81
BRITS	41.77	41.87	42.59	43.47	47.09	41.75	41.92	42.37	43.44	46.36	41.45	41.56	42.58	40.36	25.19
CSDI	41.77	41.91	42.72	43.73	48.54	41.75	41.92	42.43	43.78	47.94	42.06	43.62	43.27	44.63	25.01
Variance share of the first five (%) (true 60.02)															
KNN	60.04	60.09	60.32	60.35	61.19	60.02	60.15	60.21	60.62	60.92	59.70	58.28	59.49	57.71	–
LowRank	60.06	60.26	61.37	62.56	69.35	60.03	60.26	61.07	62.50	67.76	60.12	60.67	61.36	63.34	51.75
EM	60.06	60.22	61.19	62.05	67.16	60.03	60.24	60.88	62.14	66.09	60.07	60.14	61.03	61.77	49.58
BRITS	60.05	60.16	61.14	61.96	66.74	60.03	60.24	60.84	62.08	65.70	59.73	59.92	60.96	59.06	50.00
CSDI	60.06	60.22	61.18	62.03	67.28	60.03	60.24	60.85	62.22	66.31	60.23	61.24	61.34	62.49	49.66
Subspace distance D_1															
KNN	0.0006	0.0030	0.0082	0.0133	0.0252	0.0008	0.0034	0.0089	0.0179	0.0511	0.0106	0.0309	0.0169	0.0398	–
LowRank	0.0006	0.0026	0.0068	0.0100	0.0215	0.0004	0.0031	0.0053	0.0143	0.0375	0.0079	0.0180	0.0149	0.0268	0.1186
EM	0.0005	0.0026	0.0062	0.0094	0.0152	0.0005	0.0021	0.0047	0.0097	0.0262	0.0063	0.0178	0.0126	0.0232	0.1148
BRITS	0.0005	0.0034	0.0054	0.0100	0.0161	0.0004	0.0024	0.0046	0.0104	0.0252	0.0176	0.0203	0.0132	0.1407	0.1106
CSDI	0.0006	0.0027	0.0062	0.0122	0.0211	0.0003	0.0028	0.0040	0.0158	0.0465	0.0088	0.0316	0.0170	0.0303	0.1141
Subspace distance D_5															
KNN	0.0051	0.0143	0.0364	0.0436	0.1434	0.0040	0.0117	0.0376	0.0525	0.1365	0.0276	0.1055	0.0680	0.1131	–
LowRank	0.0036	0.0195	0.0809	0.2551	0.3914	0.0033	0.0207	0.0334	0.1892	0.3291	0.0152	0.0555	0.0714	0.3293	0.3814
EM	0.0040	0.0126	0.0440	0.1120	0.2039	0.0034	0.0117	0.0249	0.0550	0.1273	0.0145	0.0950	0.0327	0.0843	0.1343
BRITS	0.0046	0.0154	0.0347	0.1034	0.2376	0.0029	0.0115	0.0270	0.0508	0.1177	0.0317	0.0900	0.0368	0.1219	0.1184
CSDI	0.0041	0.0126	0.0323	0.1014	0.1954	0.0027	0.0100	0.0291	0.0453	0.1435	0.0160	0.1166	0.0383	0.0964	0.1340